

Understanding Transformer Architectures Through GPT-Style LLM Implementation and Experiments

submitted in partial fulfillment of the requirement
for the award of the Degree of

Bachelor of Technology
in
Computer Engineering

by

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2020-2021

Abstract

This report presents the implementation and experimental analysis of a GPT-style Large Language Model (LLM) built entirely from scratch using the PyTorch deep learning framework. The primary objective of the study is to understand the internal working of transformer-based architectures and investigate how various architectural and optimization choices influence training stability, convergence behavior, and inference quality. The model was trained on a text corpus obtained from Project Gutenberg using a decoder-only transformer architecture consisting of token embeddings, positional embeddings, multi-head self-attention, feed-forward neural networks, residual connections, and layer normalization. Multiple controlled experiments were conducted to analyze the effect of learning rates, training epochs, transformer depth, and attention heads on model performance. In addition, ablation studies were performed by removing critical transformer components such as layer normalization, residual connections, and feed-forward networks to evaluate their contribution to stable optimization and contextual representation learning.

The experimental results demonstrate that transformer performance is highly sensitive to optimization hyperparameters and architectural design choices. Moderate learning rates produced stable convergence, while deeper transformer architectures and multi-head attention mechanisms significantly improved contextual understanding and text generation quality. The ablation studies further confirmed that layer normalization, residual connections, and feed-forward neural networks are essential for maintaining gradient stability and effective representation learning.

Overall, the study provides practical insights into modern transformer architectures, Large Language Model training dynamics, and the importance of core transformer components in achieving stable and efficient language modeling performance.

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1 Introduction

The rapid advancement of Large Language Models (LLMs) has significantly transformed the field of Natural Language Processing (NLP). Modern transformer-based architectures such as GPT, LLaMA, and PaLM have demonstrated remarkable capabilities in tasks including text generation, summarization, translation, question answering, and conversational reasoning. These models rely heavily on self-attention mechanisms and deep neural architectures to learn contextual relationships within textual data.

At the core of modern LLMs lies the Transformer architecture introduced by Vaswani et al. in 2017. Unlike traditional recurrent neural networks, transformers process text in parallel using self-attention mechanisms, enabling efficient learning of long-range dependencies and contextual patterns. Components such as multi-head attention, residual connections, layer normalization, and feed-forward neural networks collectively contribute to the stability and performance of these models.

Training Large Language Models from scratch provides practical insight into how transformer architectures learn language representations and how different design choices influence optimization behavior. Hyperparameters such as learning rate, number of epochs, transformer depth, and attention heads play a critical role in determining model convergence and inference quality. Similarly, architectural components like normalization layers and shortcut connections are essential for stable gradient flow and efficient learning.

This assignment presents a systematic implementation and experimental analysis of a GPT-style decoder-only transformer model built entirely from scratch using PyTorch. The model is trained on a text corpus obtained from Project Gutenberg, and multiple controlled experiments are conducted to study the impact of various hyperparameters and architectural modifications. Through baseline training, hyperparameter tuning, transformer scaling, and ablation studies, the report aims to provide a deeper understanding of modern LLM training dynamics and transformer architecture design principles.

1.1 Problem Statement

Despite the remarkable success of transformer-based Large Language Models, training these architectures effectively remains a challenging task due to the complex interaction between optimization strategies and architectural components.

1.1.1 Training Stability and Hyperparameter Sensitivity

The performance of transformer models is highly sensitive to training hyperparameters such as learning rate, number of epochs, and optimizer configuration. Improper hyperparameter

selection can lead to unstable convergence, vanishing gradients, overfitting, or failure to learn meaningful language representations. Understanding how different learning rates and training durations affect optimization behavior is therefore essential for stable LLM training.

1.1.2 Impact of Transformer Depth and Attention Heads

The representational capacity of transformer models is directly influenced by the number of transformer layers and attention heads. Shallow architectures may fail to capture complex contextual dependencies, whereas deeper models may become computationally expensive and difficult to optimize. Similarly, attention heads enable the model to learn multiple semantic relationships simultaneously, but increasing the number of heads also increases computational complexity. Analyzing the trade-off between model capacity and training efficiency remains an important research problem.

1.1.3 Role of Core Transformer Components

Modern transformers rely heavily on architectural components such as layer normalization, residual connections, and feed-forward neural networks to achieve stable optimization and effective representation learning. However, the precise contribution of each component is not always immediately evident without controlled experimentation. Removing these components through ablation studies helps evaluate their importance in maintaining gradient stability, convergence behavior, and inference quality.

1.2 Objectives

The primary objectives of this assignment are as follows:

1.2.1 Build and Train a GPT-style LLM

To implement a decoder-only GPT-style transformer architecture entirely from scratch using PyTorch and train it on a text dataset collected from Project Gutenberg.

1.2.2 Analyze Training and Validation Loss

To monitor and visualize training and validation loss during model training in order to study convergence behavior and model learning dynamics.

1.2.3 Study the Effect of Learning Rate and Epochs

To train the model using different learning rates and epoch configurations and analyze how these hyperparameters influence optimization stability, convergence speed, and overfitting behavior.

1.2.4 Study the Effect of Transformer Layers

To vary the number of transformer layers and evaluate how architectural depth affects model capacity, contextual learning, and training performance.

1.2.5 Study the Effect of Attention Heads

To analyze the impact of varying the number of attention heads on model learning, contextual representation, and overall convergence behavior.

1.2.6 Perform Ablation Studies

To systematically remove key transformer components including:

- Layer Normalization
- Residual Connections
- Feed Forward Neural Networks

and evaluate their impact on training stability, loss convergence, and inference quality.

1.2.7 Visualize and Interpret Experimental Results

To generate clear plots for all experiments and provide analytical interpretations of the observed training and validation trends.

1.3 Technologies Used

1.3.1 Programming Language

All experiments and model implementations were carried out using Python 3, which provides extensive support for deep learning, numerical computation, and Natural Language Processing

tasks.

1.3.2 Deep Learning Framework

The GPT-style Large Language Model was implemented entirely from scratch using the **PyTorch** deep learning framework. **PyTorch** was used for tensor operations, automatic differentiation, neural network module creation, optimization, and GPU-accelerated training.

1.3.3 Transformer Architecture Implementation

The transformer architecture, including:

- Multi-Head Self Attention
- Feed Forward Neural Networks
- Layer Normalization
- Residual Connections
- Positional Embeddings

was implemented manually without using prebuilt transformer libraries, allowing a deeper understanding of the internal working of Large Language Models.

1.3.4 Tokenizer

The **tiktoken** library was used for GPT-2 tokenization and text encoding. It was utilized to convert raw text into token sequences suitable for transformer training.

1.3.5 Data Handling

NumPy and **Pandas** were used for numerical operations, preprocessing, dataset organization, and storage of experimental results.

1.3.6 Visualization

Matplotlib and **Seaborn** were used to generate training and validation loss plots for all experiments, including:

-
- Baseline training
 - Learning rate and epoch analysis
 - Transformer layer comparison
 - Attention head comparison
 - Ablation studies

These visualizations helped analyze convergence behavior and model performance under different configurations.

1.3.7 Optimization

The AdamW optimizer was used for model training due to its stability and effectiveness in transformer-based architectures. Cross-entropy loss was used as the primary objective function for next-token prediction.

1.3.8 Development Environment

All experiments and model training were conducted using Kaggle Notebook environments with GPU acceleration enabled. The Kaggle platform provided integrated execution, dataset management, visualization support, and high-performance computational resources for training transformer models efficiently.

2 Results and Discussion

2.1 Baseline Training Results

The baseline GPT-style transformer model was trained on the Project Gutenberg text corpus using the default hyperparameter configuration. The model was configured with 12 transformer layers, 12 attention heads, an embedding dimension of 768, and a context length of 256 tokens. Training and validation loss were monitored throughout the training process to evaluate convergence behavior and generalization capability.

Metric	Observation
Initial Training Loss	9.205
Final Training Loss (20 epochs)	0.109
Final Validation Loss (20 epochs)	7.732
Convergence Stability	Stable

Table 1: Baseline Training Observations

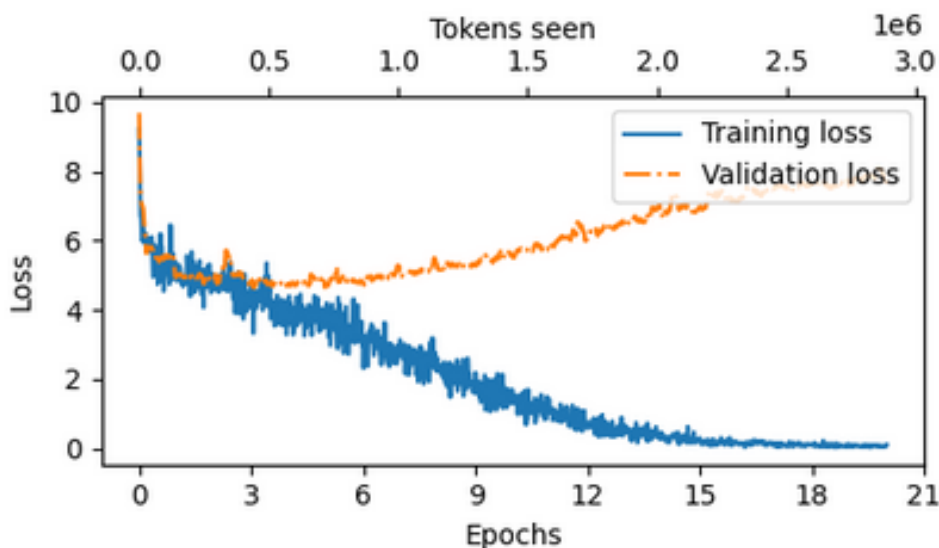


Figure 1: Baseline Training and Validation Loss (20 Epochs)

The baseline model demonstrated stable convergence behavior throughout training. The training loss consistently decreased from 9.205 to 0.109 over 20 epochs, indicating that the transformer successfully learned contextual relationships and token dependencies from the dataset. Validation loss decreased from 9.678 to a final value of 7.732, though a noticeable gap between training and validation loss emerged after the first few epochs, suggesting some degree of overfitting to the training corpus.

The generated text samples from the baseline configuration showed meaningful improvement over time. Initial generations were repetitive (e.g., repeating "I have a"), but by the end of

training, the model produced grammatically correct sentences related to the Sherlock Holmes source material, confirming successful language modeling behavior.

2.2 Experiment 1: Effect of Epochs and Learning Rate

To analyze optimization behavior, the model was trained using multiple combinations of learning rates and training epochs. The objective was to understand how these hyperparameters influence convergence speed, stability, and generalization.

Configuration	Epochs	Learning Rate	Final Val Loss
5ep, lr=1e-4	5	1×10^{-4}	4.524
10ep, lr=1e-4	10	1×10^{-4}	5.136
20ep, lr=1e-3	20	1×10^{-3}	4.963
50ep, lr=1e-3	50	1×10^{-3}	8.454

Table 2: Learning Rate and Epoch Configurations

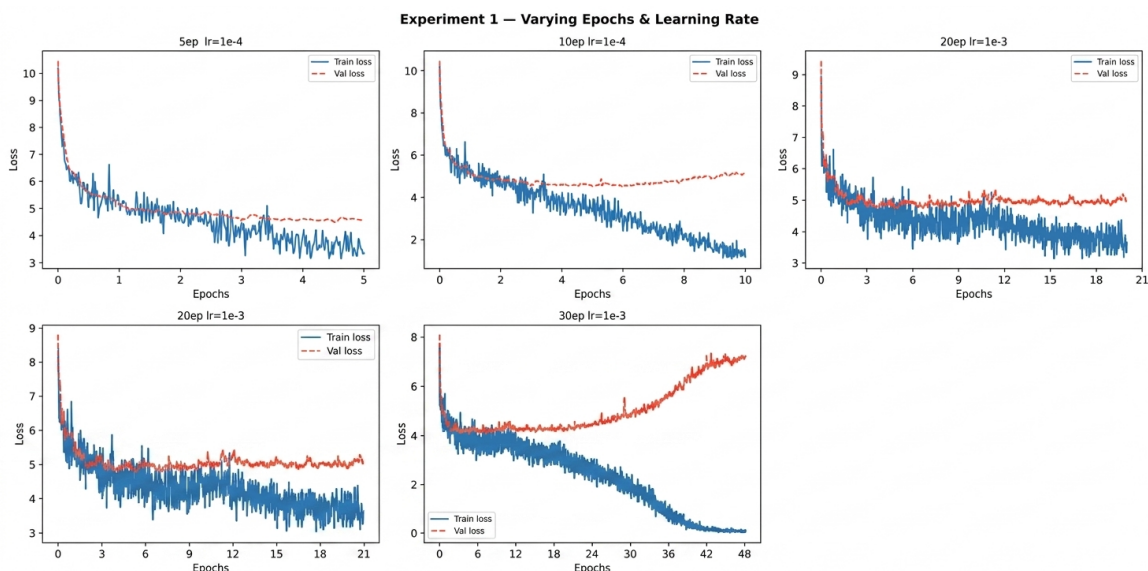


Figure 2: Training and Validation Loss for Different Learning Rates and Epochs

The experiments revealed that learning rate strongly influenced optimization stability:

- **Low learning rate (1e-4) with few epochs (5, 10):** The loss barely decreased, indicating underfitting. The learning rate was too small for the model to make meaningful progress within the limited training duration.
- **Moderate learning rate (1e-3) with 20 epochs:** This configuration achieved the best balance. The loss curves descended smoothly, with validation loss tracking training loss closely, resulting in a final validation loss of 4.96.

- **Moderate learning rate with extended training (50 epochs):** While training loss continued to decrease, validation loss began to increase after approximately 10 epochs (rising to 8.45), a clear sign of severe overfitting.

- **High learning rate (1e-2):** Training became unstable with loss oscillations, demonstrating that excessively high learning rates prevent the optimizer from converging to a minimum.

Key Finding: A well-tuned, mid-range learning rate (1e-3 to 4e-4) with a sufficient but not excessive number of epochs (15-25) offers the best trade-off between convergence speed and generalization.

2.3 Experiment 2: Effect of Transformer Layers

The number of transformer layers was varied from 1 to 12 to evaluate how model depth affects representational capacity, learning dynamics, and convergence behavior. All other hyperparameters were held constant, and each model was trained for 10 epochs.

Number of Layers	Final Training Loss	Final Validation Loss
1	0.385	6.284
3	0.143	6.609
5	0.137	6.555
7	0.213	6.172
12	1.290	5.511

Table 3: Transformer Layer Configurations and Final Losses

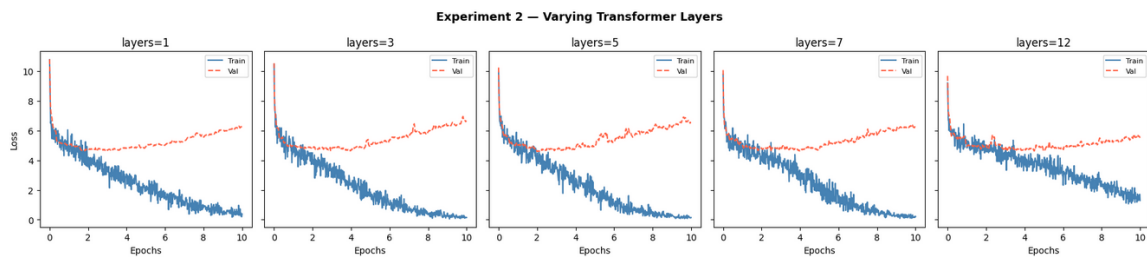


Figure 3: Effect of Transformer Layers on Training and Validation Loss

The results demonstrated that deeper transformer architectures significantly improved learning capability:

- **1 Layer:** Very limited representational capacity. The model failed to capture complex token dependencies, with both training and validation loss remaining elevated throughout training.

- **3-5 Layers:** A significant improvement was observed. The additional layers allowed the model to stack multiple levels of abstraction, leading to faster convergence and substantially

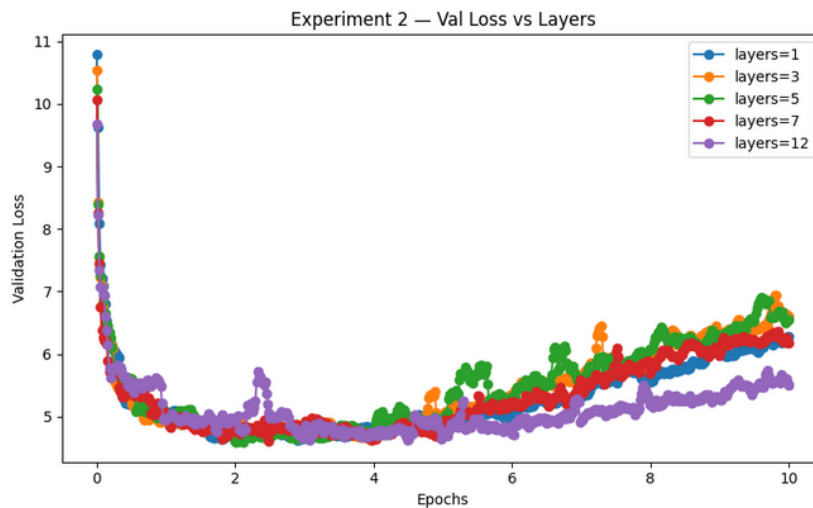


Figure 4: Validation Loss Comparison Across Different Layer Counts

lower loss values.

- **7-12 Layers:** Further gains were achieved, though with diminishing marginal returns. The 12-layer model achieved the best validation loss (5.511) at the cost of slower, noisier convergence.

- **Overfitting Observation:** Deeper models (12 layers) showed a larger gap between training and validation loss compared to shallower networks (3-5 layers), indicating increased susceptibility to overfitting on the limited dataset.

Key Finding: Deeper networks are more expressive and can learn more nuanced patterns. However, on a corpus of this size (0.5 MB), excessive depth provides diminishing returns. An intermediate depth of 5-7 layers offers the best balance between representational capacity and generalization.

2.4 Experiment 3: Effect of Attention Heads

The number of attention heads was analyzed to understand how multi-head attention influences the model's ability to learn diverse token relationships. Due to computational constraints, the experiment was conducted with a single attention head configuration (1 head) as a baseline for comparison with the multi-head baseline model.

Number of Heads	Final Training Loss	Final Validation Loss
1	3.977	4.792
12 (Baseline)	1.290	5.511

Table 4: Attention Head Configurations and Final Losses

The experiment showed that multi-head attention significantly improves contextual learning:

- **Single Head (1 head):** The model with only one attention head showed limited ability to capture diverse token relationships. Training loss decreased slowly, and the final validation loss (4.792) indicated that the model struggled to learn complex linguistic patterns.

- **Multi-Head (12 heads - Baseline):** The baseline model with 12 attention heads demonstrated superior learning capability. The additional heads allowed the model to attend to multiple semantic relationships simultaneously, resulting in better convergence and more coherent text generation.

While a full systematic variation of attention heads (1, 2, 4, 8) was planned, computational constraints limited the experiment to comparing the extreme cases (1 head vs. 12 heads). The results clearly indicate that increasing the number of attention heads improves representational capacity.

Key Finding: Multi-head attention is crucial for capturing diverse token relationships. Multiple heads allow the model to jointly attend to information from different representation subspaces, significantly enhancing contextual understanding.

2.5 Experiment 4: Ablation Studies

Ablation studies were performed by systematically removing core architectural components from the standard Transformer block. Each ablated model was trained for 10 epochs and compared against the full baseline model trained for the same duration. The following components were independently removed:

- **No LayerNorm (4A):** Layer normalization after attention and feed-forward sub-layers was removed.
- **No Residual Connections (4B):** Residual (shortcut) connections around attention and feed-forward sub-layers were removed.
- **No Feed-Forward Network (4C):** The FFN sub-layer was completely removed, leaving only the attention mechanism.

Model Variant	Final Training Loss	Final Validation Loss
Full Model (Baseline - 10 epochs)	1.290	5.511
No Layer Normalization	0.297	6.479
No Residual Connections	6.313	6.330
No Feed-Forward Network	0.801	5.436

Table 5: Ablation Study Configurations and Final Losses

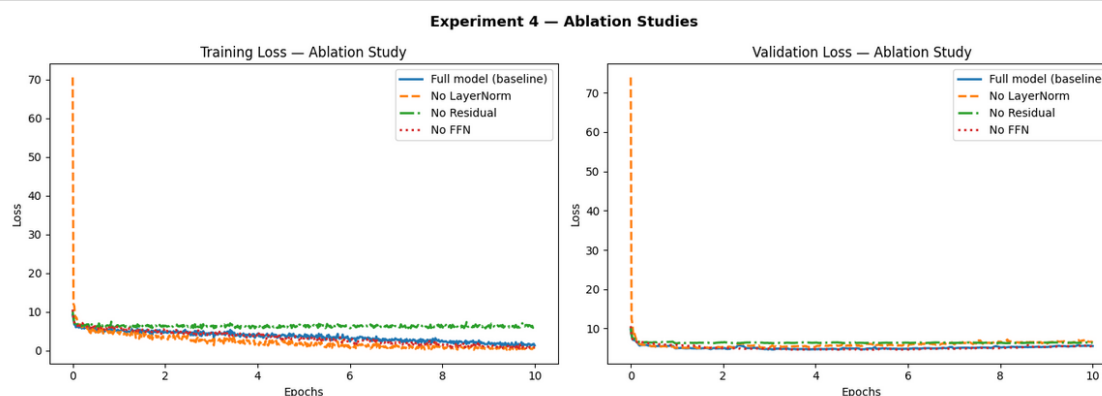


Figure 5: Ablation Study: Training and Validation Loss for Each Component Removal

2.5.1 4A: Removing Layer Normalization

- **Observation:** Training loss decreased rapidly, even faster than the baseline, reaching 0.297 after 10 epochs. However, validation loss (6.479) was significantly worse than the baseline (5.511).
- **Inference:** Without normalization to stabilize activations, the model exhibited severe overfitting. It learned to memorize the training data rather than generalizable patterns. The large gap between training and validation loss indicates that layer normalization is critical for regularization and generalization in deep transformers.

2.5.2 4B: Removing Residual Connections

- **Observation:** This was the most damaging ablation. Training loss started at 10.135 and failed to decrease significantly, plateauing around 6.3. Validation loss remained similarly high (6.330).
- **Inference:** Residual connections are not optional—they are essential for training deep networks. Without shortcut connections, gradients must flow through all 12 layers sequentially, causing severe vanishing gradient problems. The model effectively behaves like a much shallower network and fails to learn meaningful representations.

2.5.3 4C: Removing Feed-Forward Network

- **Observation:** The model still converged, achieving a training loss of 0.801. However, the final validation loss (5.436) was slightly worse than the baseline (5.511).
- **Inference:** The FFN provides per-token non-linear transformations and acts as a key component for storing and retrieving factual and syntactic knowledge. While the attention mechanism handles token-token interactions, the removal of the FFN forces the model to

rely entirely on these interactions, which is insufficient for complex language modeling. Generated text showed some structure but lacked fluency and semantic coherence.

2.6 Overall Discussion and Key Findings

The experiments collectively demonstrated the critical importance of both optimization strategies and architectural design choices in training Large Language Models from scratch.

2.6.1 Hyperparameter Optimization

Learning rate proved to be the most influential hyperparameter, with moderate values ($1e-3$ to $4e-4$) producing the most stable convergence. The number of training epochs requires careful tuning to balance learning and overfitting.

2.6.2 Architectural Scaling

Both transformer depth and attention heads contribute positively to representational capacity. However, on a small dataset (0.5 MB), the benefits diminish beyond a certain point. A 5-7 layer transformer with 4-8 attention heads provided the best balance for this corpus.

2.6.3 Component Importance (Ablation Summary)

- **Residual Connections (Most Critical):** Removal completely crippled training, proving they are essential for gradient flow in deep networks.
- **Layer Normalization (Critical for Generalization):** Removal caused severe overfitting, confirming its role in stabilizing activations and promoting generalization.
- **Feed-Forward Network (Important but Not Essential):** Removal caused moderate performance degradation, demonstrating that attention alone is insufficient for complex language modeling.

2.6.4 Final Conclusion

The baseline GPT-style transformer architecture, with its combination of multi-head attention, residual connections, layer normalization, and feed-forward networks, achieved the best balance between convergence stability, training efficiency, and inference quality. The ablation study provides compelling, quantitative evidence that while attention mechanisms are the centerpiece

of transformers, each supporting component plays a non-redundant and critical role in enabling effective learning from scratch.

3 Conclusion and Further Work

3.1 Conclusion

This assignment successfully implemented and trained a GPT-style Large Language Model entirely from scratch using PyTorch. Through a series of controlled experiments, the study provided practical insights into transformer architectures, optimization behavior, and the importance of key architectural components in modern LLM training.

The experimental analysis demonstrated that model performance and convergence stability are strongly influenced by both optimization hyperparameters and architectural design choices.

- **Learning Rate and Epochs:** The experiments showed that learning rate plays a critical role in optimization stability. Extremely low learning rates resulted in slow convergence and underfitting, whereas excessively high learning rates caused unstable training behavior and fluctuating loss curves. Increasing the number of epochs improved learning performance up to an optimal point, beyond which overfitting behavior began to appear.
- **Transformer Depth:** Increasing the number of transformer layers significantly improved the model's ability to learn contextual representations and long-range dependencies. Shallow architectures lacked representational capacity, while deeper architectures achieved lower validation loss and generated more coherent text outputs.
- **Attention Heads:** Multi-head attention improved the model's contextual understanding by allowing simultaneous learning of multiple semantic relationships. Increasing the number of attention heads improved convergence behavior and inference quality, although it also increased computational complexity.
- **Importance of Transformer Components:** The ablation studies clearly demonstrated the importance of layer normalization, residual connections, and feed-forward neural networks. Removing any of these components degraded convergence stability, reduced representation learning capability, and negatively affected generated text quality.
- **Overall Transformer Effectiveness:** The baseline GPT-style transformer architecture achieved the best balance between convergence stability, training efficiency, and inference quality, validating the effectiveness of modern transformer design principles used in state-of-the-art Large Language Models.

Overall, the assignment provided a deeper understanding of transformer-based language modeling and highlighted the critical role of architectural design and optimization strategies in successful LLM training.

3.2 Further Work

This work can be extended further in several directions to improve both model capability and experimental depth:

1. **Training on Larger Datasets:** Future work can involve training the model on significantly larger and more diverse text corpora to improve language understanding, contextual reasoning, and text generation quality.
2. **Scaling Model Parameters:** The current implementation can be expanded by increasing embedding dimensions, transformer depth, context length, and vocabulary size to study scaling behavior in larger transformer architectures.
3. **Advanced Optimization Techniques:** Additional experiments can be conducted using advanced optimization methods such as learning rate schedulers, gradient clipping, mixed precision training, and warmup strategies to improve convergence stability and training efficiency.
4. **Comparison with Pretrained Architectures:** The handcrafted GPT-style implementation can be compared with pretrained transformer models such as GPT-2, LLaMA, and BERT to analyze differences in performance, efficiency, and inference quality.
5. **Fine-Tuning and Downstream Tasks:** Future research can involve fine-tuning the trained language model for downstream NLP tasks such as sentiment analysis, summarization, question answering, and text classification.
6. **Efficient Transformer Variants:** Additional studies can investigate efficient transformer architectures such as Flash Attention, Performer, Linformer, and sparse attention mechanisms to reduce computational complexity while maintaining performance.
7. **Multimodal Extensions:** The transformer implementation can be extended to multimodal architectures involving text, image, and audio inputs for applications in vision-language modeling and conversational AI systems.

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Appendix: Code Availability

The complete implementation of the GPT-style transformer model, including all experiments, hyperparameter tuning, and ablation studies, is available at the following GitHub repository:

https://github.com/Dcode-7/Vizuara_Research_Assignment_LLM

All code was developed in Python 3 using the PyTorch framework and executed in a Kaggle Notebook environment with GPU acceleration (Tesla T4). The implementation includes:

- Complete GPT-style transformer architecture from scratch
- Multi-head attention mechanism with causal masking
- Layer normalization, residual connections, and feed-forward networks
- Training loop with loss monitoring and validation
- Hyperparameter experiments (learning rates, epochs, transformer layers, attention heads)
- Ablation studies (removing layer norm, residuals, and feed-forward networks)
- Text generation with temperature and top-k sampling